

CHLOÉ SERRE-COMBE

PhD in Statistics · Extreme and spatio-temporal statistics

IMAG, University of Montpellier, France

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EDUCATION

PhD in Statistics

2022–2026

IMAG, University of Montpellier and Inria LEMON team, France

Master's degree in Statistics and Data Science, Biostatistics specialization

2020–2022

University of Montpellier, France

EXPERIENCE

Temporary Lecturer and Research Assistant (ATER)

2025–2026

IMAG, University of Montpellier and Inria LEMON team

Montpellier, France

PhD Candidate with Teaching Duties

2022–2026

IMAG, University of Montpellier and Inria LEMON team

Montpellier, France

- Thesis topic: Stochastic generator for extreme precipitation in urban areas at high spatio-temporal resolution
- Methods: Spatio-temporal extreme-value modeling, extreme dependence and statistical downscaling
- Data: OMSEV (in situ), COMEPHORE (radar reanalysis)
- Supervisors: Nicolas MEYER, Thomas OPITZ and Gwladys TOULEMONDE

Data Science Apprentice

September 2021 – August 2022

KanopyMed

Clapiers, France

- Master thesis: Quantification and modeling of mortality related to air pollution
- Methods: Spatial statistics, epidemiology
- Data: ATMO, CHIMERE, INSEE, Public Health France, ERA5
- Supervisors: Ulysse RODTS and Grégoire MERCIER

Intern

July – August 2021

IMAG, University of Montpellier

Montpellier, France

- Enhanced the RKeOps package, a software for 2D matrix reduction computations on GPUs
- Supervisors: Benjamin CHARLIER and Ghislain DURIF

Intern

June – July 2018

Aix-Marseille University

Aix-en-Provence, France

- Miller–Rabin primality test: understanding, oral presentation and report writing
- Supervisor: Julien KELLER

PUBLICATIONS

PREPRINT

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Submitted, 2026. Preprint: arXiv:2602.19774.

CONFERENCE PROCEEDINGS

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Journées de Statistique (JDS), 2025.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. EVAN, 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. Journées de Statistique (JDS), 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modélisation statistique de précipitations urbaines à fine échelle spatio-temporelle*. Journées de Statistique (JDS), 2023.

ORAL PRESENTATIONS

INVITED SEMINARS

- May 2026** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
KIT, Karlsruhe, Germany
- Feb. 2026** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
J.A. Dieudonné Laboratory, Nice, France
- Jan. 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
PhD Seminar, IMAG, University of Montpellier, France

INTERNATIONAL CONFERENCES

- Dec. 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
SWGEM, Grenoble, France
- April 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
EGU, Vienna, Austria
- July 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVAN, Istituto Veneto di Scienze, Lettere ed Arti, Venice, Italy
- June 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
IMSC, Météo France, Toulouse, France
- June 2023** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVA, Bocconi University, Milan, Italy

NATIONAL CONFERENCES AND SEMINARS

- June 2026** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Rencontres Montpellier–Sherbrooke, IMAG, University of Montpellier, France
- June 2025** *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Journées de Statistique (JDS), Aix-Marseille University, Marseille, France
- Dec. 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Occimath Seminar, IMT, Toulouse, France
- May 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Journées de Statistique (JDS), University of Bordeaux, France
- April 2024** *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Rencontres des Jeunes Statisticien.ne.s (RJS), Porquerolles, France
- Sept. 2023** *Modeling urban precipitation at high spatio-temporal resolution*
MUFFINS Workshop, Aix-en-Provence, France
- July 2023** *Modeling urban precipitation at high spatio-temporal resolution*
Journées de Statistique (JDS), University of Brussels, Belgium
- June 2023** *Towards statistical modeling of urban rainfall on a fine spatio-temporal scale*
RESSTE Day, FRUMAM, Marseille, France

TEACHING

Scientific reasoning (54 h)

2026

1st year Bachelor's degree in Biology, University of Montpellier, France

- Introduction to probability
- Lectures, tutorials and exam grading

Mathematics for PeiP (108 h)

2025–2026

1st year of the integrated preparatory programme (PeiP), Polytech Montpellier, France

- First-year algebra and analysis
- Tutorials and exam grading

Computational methods (21 h)

2025

1st year Bachelor's degree in Biology, University of Montpellier, France

- Introduction to mathematics for biology, functions, proportionality...
- Tutorials and exam grading

Modeling and Statistics (101 h)

2022–2025

3rd year engineering programme, Computer Science and Management specialization, Polytech Montpellier, France

- Descriptive statistics, conditional probability, discrete and continuous random variables, random vectors and independence, R
- Lectures, tutorials, practical sessions in R, exam preparation and grading

Probability, Modeling and Statistics (19 h)

2024

3rd year engineering programme, Software Development and Operations specialization, Polytech Montpellier, France

- Conditional probability, discrete and continuous random variables
- Lectures, tutorials, exam preparation and grading

Multivariate Data Analysis (12 h)

2023

4th year engineering programme, Computer Science and Management specialization, Polytech Montpellier, France

- Statistics, linear regression, data visualization, R, ggplot, Shiny
- Online course available at github.com/chloesrcb/ig4-dataviz, practical sessions

Linear Algebra (27 h)

2023

1st year Bachelor's degree, Physics–Chemistry–Engineering Sciences track (PCSI), University of Montpellier, France

- Linear systems, vector spaces, matrices, linear maps, diagonalization, change of basis...
- Tutorials and exam grading

SKILLS

Climate data: NetCDF, radar data, in situ observations, reanalysis products

Programming: Python, R, Bash

Tools: Git, LaTeX, VS Code, HPC environments, QGIS (basic)

Operating systems: Linux, Windows

Languages: French (native), English (fluent), Italian (intermediate)

INVOLVEMENT

Co-organizer of a working group on extreme value statistics

2024–2026

IMAG, University of Montpellier, France

Organization of a working group among PhD students, meeting every two to three months to discuss research articles in extreme value statistics.

Co-organizer of the Mathematics PhD seminar

2023–2024

IMAG, University of Montpellier, France

Organization of a biweekly doctoral seminar, including inviting speakers and managing the budget. Organization of a PhD research day with poster sessions.

Outreach activity for undergraduate women in mathematics (*Maths C pour L*)

Feb. 2024

CIRM, Marseille, France

Presentation of my academic path and research activities, followed by discussions with students to promote careers in science and mathematics.